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Traceable Decisions Without Backpropagation: Graph Concept Attractors for Handwritten Digit Recognition

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***Abstract*—Post-hoc explanation methods approximate a trained model from the outside, and a model can be built so that the explanation hides the behaviour it is meant to expose. We build a recogniser whose decision is already an inspectable object. Each grey-scale image is converted into an attributed bipartite graph: Point nodes mark the endpoints, corners and intersections of the contour skeleton, and Vector nodes mark the straight segments between them. For every class we form a concept attractor by structural reduction, merging training graphs one at a time and generalizing attribute values to intervals. No gradient is computed and no backpropagation is used. A test image is classified by graph edit distance against every concept attractor, with a winner-take-all rule. We evaluate on the MNIST test images whose contours are complete: the completeness filter admits 8,707 of 10,000, and the headline metrics are conditional on it. Thirteen concept attractors formed from 76 unique training samples (805 images after augmentation) classify 8,685 images with 91.13% accuracy, 91.92% precision, 91.13% recall and 91.34% F1 score. Counted over all 10,000 test images, the system assigns a class to 86.41% of them at a selective risk of 8.40%; run with no filter, it classifies 82.21% of the test set correctly. We trace one decision from the input image through the winning concept attractor down to the per-feature costs. End-to-end inference over the evaluation set took 147 to 232 ms per image on one workstation. We report a research prototype and make no comparison with neural classifiers or with other structural methods.**

***Index Terms*—explainable artificial intelligence, graph-based image representation, concept attractor, structural reduction, graph edit distance, traceable decision**

I. Introduction

Recognition systems reach high accuracy and return a label without showing the evidence behind it. Where a decision has to be justified to an operator, a clinician or an auditor, that opacity is a separate problem from how often the model is correct.

The usual remedy is a post-hoc explanation. LIME builds a local linear approximation of a trained model around one example [1]. SHAP distributes feature contributions using a cooperative-game scheme [2]. Both work from outside the model and treat it as a black box, so the explanation describes a surrogate function. Slack et al. showed that the gap can be exploited: a classifier with an explicit bias can be built so that LIME and SHAP hide that bias [3]. Rudin argues that high-stakes decisions call for a model interpretable by construction [4]. We take that position.

In our system the structure of the representation carries the explanation. The contour of an image is encoded as an attributed graph whose nodes are characteristic points and the segments between them, with geometric and semantic attributes stored on the nodes themselves. A cycle in the graph corresponds to a closed contour, node degree describes branching, and an attribute interval states how far that feature may vary inside a class. We build no explanation module on top of the classifier.

The system is a prototype built for research. It does not claim the accuracy of neural classifiers on benchmark data and we do not compare it against them. Our question is whether recognition is possible without backpropagation, from a small number of training samples, with a decision path a reader can follow from end to end.

The paper contributes: (i) a graph representation and a concept formation procedure that use no gradient and no backpropagation; (ii) an evaluation on MNIST under a completeness filter we state in full, with its per-class skew and with figures over the full test set; (iii) a worked decision trace that explains one correct decision and the dominant error; and (iv) inference cost measured end to end and per pipeline stage on run artefacts.

II. Related Work

Baniecki and Biecek survey attacks on explanations: explanations can be manipulated, fooled or fairwashed, and the survey catalogues the resulting insecurities [5]. The alternative design direction is a model interpretable by construction [4]. Forest et al. apply concept bottleneck models to prognostics, taking degradation modes as the intermediate concepts so a prediction decomposes per component [6]; the mapping from input to concept is still a trained network. García-Cuesta et

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al. show that the prototypes of prototype-based networks can be altered while the explanation keeps its plausibility [7]. Interpretability by construction therefore needs representation elements with a geometric meaning outside any learned embedding space.

Structural representations of shape are long established. Vision GNN treats an image as a graph of patch nodes and runs message passing over it [8]. Chatbri et al. compare contours and skeletons for matching binary images [9]; Shen et al. combine the two, recognising shapes from a bag of skeleton-associated contour parts [10]. Gan et al. represent handwritten Chinese characters by skeleton graphs and run a graph transformer over them [11]; by choice of representation this is the closest prior work to ours. There the network weights determine the decision; here the image graph is matched against a concept attractor and the answer is a list of node substitutions with their costs.

Conte et al. survey thirty years of exact and approximate graph matching [12]. Graph edit distance (GED) gives a dissimilarity that decomposes into elementary operations on nodes and edges. Riesen and Bunke approximate GED by bipartite assignment, which made it usable at practical graph sizes [13]. Wang et al. learn the A* search heuristic for GED and recover the edit path [14], and Piao et al. recover the edit path from a predicted matching matrix, noting that models which reduce GED to similarity regression produce no operation sequence [15]. In formulations that regress similarity directly, an elementary operation has no predefined cost, so a single cost term does not say which feature separates two matched nodes.

Work on alternatives to backpropagation continues. The Forward-Forward algorithm replaces the backward pass with two forward passes and a local objective [16], and a survey of neuro-symbolic integration describes systems in which an explicit symbolic part carries explainability and formal reasoning [17]. Few-shot learning is dominated by meta-learning: prototypical networks build a class prototype as a centroid in a learned embedding space [18], MAML learns initial weights that a few gradient steps adapt to a new task [19], and Lake et al. learn a character from one example with a generative program model [20]. We cite these as context for the problem setting and make no accuracy comparison against them. The paper continues our own graph model of few-shot learning without backpropagation [21] and the information-architecture frame behind it [22].

The reviewed work leaves three conditions unmet at once. The structure of the representation should carry both the decision and the explanation. Matching costs should be tied to predefined geometric properties instead of being learned from data. The result should come from tens of training samples without backpropagation. This paper addresses that combination.

III. Method

A. Graph Representation

We represent a digit image by an attributed bipartite graph whose two kinds of node alternate: a Point node is a characteristic point of the contour, a Vector node the straight segment between two neighbouring points. Links are bidirectional, so every traversal has the form Point $\rightarrow$ Vector $\rightarrow$ Point; no two Point nodes are joined directly. Pixel values are not used after the graph is built. Making a segment a node rather than an edge gives it attributes of its own — length, direction, orientation — so it takes part in comparison like a point.

Point nodes have four types: an EndPoint terminates an open contour, a CornerPoint marks a sharp change of direction and stores the angle, an IntersectionPoint marks a crossing of lines, and a StartPoint fixes where traversal begins. One node can carry several of these labels at once.

A Point node stores its x and y coordinates and their normalized values. A Vector node stores both endpoints, the length, the quadrant (I–IV, or -1 for an axis-aligned segment), the horizontal direction (LEFT, RIGHT, NONE) and the vertical direction (TOP, BOTTOM, NONE); quadrant and directions describe orientation without absolute coordinates. Coordinates are normalized to a centred range from -1 to 1 by (1):

$$x_{norm} = \frac{x - c_x}{\rho}, \qquad y_{norm} = \frac{y - c_y}{\rho} \tag{1}$$

where (c_x, c_y) is the centre of the bounding box of the graph's points and $\rho = \sqrt{h_w^2 + h_h^2}$ its half-diagonal, built from the half-width h_w and the half-height h_h; the denominator is the same for both axes and values are rounded to one decimal. Because the centre and the scale come from the graph's own bounding box, the normalized coordinates do not depend on where the digit sits in the image or how large it is drawn.

B. Image-to-Graph Pipeline

MNIST images are grey-scale and are upscaled from 28×28 to 100×100 pixels by Lanczos resampling. The graph is built in six stages (Fig. 1). The image is binarized at $\theta = 110$. The binary mask is cleaned by small-object removal and morphological closing, then thinned to one-pixel width by the Zhang–Suen algorithm [23]; skeleton branches shorter than 8% of the total skeleton length are pruned. The skeleton is approximated by a growing neural gas [24], whose network of nodes reproduces the topology of the skeleton lines. That network is simplified by the Ramer–Douglas–Peucker algorithm with $\varepsilon = 4.55$ [25], which drops nodes lying almost on a straight line. The bipartite graph of Point and Vector nodes is built on the simplified network, and the last stage computes node attributes and normalizes coordinates.

$\theta = 110$ is the starting threshold of a retry ladder: if the resulting skeleton graph is disconnected, the pipeline re-runs at $\theta - 5$, down to 20; only $\varepsilon = 4.55$ is fixed. The same pipeline runs on training images during concept formation and on test images during classification.

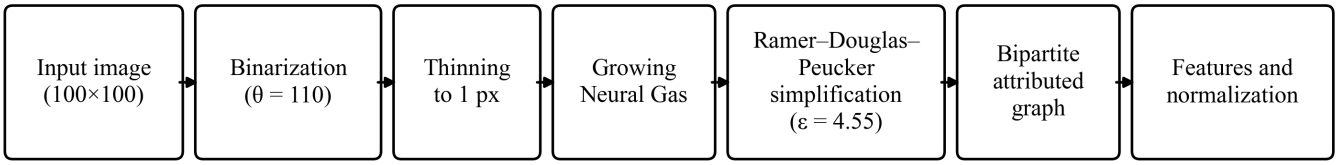


Fig. 1. The six stages that turn a 100×100 grey-scale image into an attributed bipartite graph: binarization at $\theta = 110$, thinning to one pixel, growing neural gas, Ramer–Douglas–Peucker simplification at $\varepsilon = 4.55$, construction of the Point and Vector nodes, and attribute extraction with coordinate normalization.

Require: sample graphs $G_1, \ldots, G_n$ of one concept
derive the start-point rule from all G_j and mark the StartPoint of each
$C \leftarrow G_1$
for $j = 2, \ldots, n$ **do**
  $A \leftarrow C$; $B \leftarrow G_j$; $r \leftarrow 0$
  while the critical graphs of A and B are not isomorphic **do**
    if $r = 5$ **then** fail
    apply endpoint reduction to (A, B)
    apply intersection reduction to (A, B)
    apply corner reduction to (A, B); $r \leftarrow r + 1$
  end while
  $S \leftarrow$ synchronous traversal of A and B from the StartPoints
  $C \leftarrow$ empty graph
  for each two consecutive matched critical points in S **do**
    take the most similar pair of paths joining them
    align the inner nodes one to one, in path order
    add each aligned pair to C with merged attributes
  end for
  if C is empty **then** fail
end for
return C

Fig. 2. Formation of one concept attractor. Nodes of C keep the identifiers of the concept side, A.

C. Concept Formation by Structural Reduction

A concept attractor is formed iteratively, one training sample at a time. Its initial value is the graph of the first sample, $C_0 = G_1$, and each further sample refines it by a custom reduction operation (CRO), (2):

$$C_{i+1} = \mathrm{CRO}(C_i, G_{i+1}), \quad i = 1, \ldots, n-1 \tag{2}$$

where C_i is the current state, G_{i+1} the graph of the next sample and n the number of samples. Fig. 2 gives the procedure. One CRO step reduces the two graphs until their critical graphs are isomorphic, traverses both synchronously, and merges the properties of corresponding nodes. The critical graph keeps the nodes labelled StartPoint, EndPoint, IntersectionPoint or CornerPoint and joins two of them when a path of other nodes connects them; isomorphism requires equal label sets. The result is a maximum common minor of the two graphs, so a stroke of three nodes in one sample can match five in another. The reduction leaves the structure that repeats in every sample of the class.

The start-point rule is derived once from all samples of the concept. If every sample graph has a node of degree one, the candidates are EndPoints; otherwise they are CornerPoints and IntersectionPoints. OPTICS clustering groups the candidates of all samples by normalized position. A cluster qualifies when at least 90% of the samples contribute to it. Among qualifying clusters of a closed class an intersection cluster is preferred to a corner cluster, and the cluster with the smallest $2\bar{x} + \bar{y}$ then wins, where $(\bar{x}, \bar{y})$ is its mean position. In each sample the cluster member nearest the cluster centroid becomes the StartPoint.

Each round applies three reduction strategies in a fixed order. Endpoint reduction compares the EndPoints of the two graphs by a distance that weights position at 0.7, stroke direction at 0.15 and branch corner density at 0.15. It removes an EndPoint whose nearest counterpart is farther than 0.31, unless the two are mutual nearest neighbours; if the counts still differ, it removes the surplus EndPoints with the largest assigned distance. Removal deletes the EndPoint and the path back to the nearest intersection. Intersection reduction relabels an IntersectionPoint of degree two as a CornerPoint and one of degree one as an EndPoint. If the counts still differ, the graph with more loses the surplus IntersectionPoints whose best similarity to the other graph's intersections is lowest. Corner reduction matches the sub-paths between non-corner critical points. Where two matched sub-paths differ in corner count, an order-preserving minimum-distance matching keeps the corners of the longer one that pair with the shorter, and the rest lose the CornerPoint label. At inference a concept node matches an image node only if the concept node's labels are a subset of the image node's labels, so a concept attractor is never more specific than the images it matches. StartPoint is never reduced; it fixes the entry point of traversal.

The traversal walks both reduced graphs breadth-first from their StartPoints over critical points. At an intersection it pairs the outgoing branches by the type of the critical point each leads to and, within one type, by the distance between those points. Between two consecutive matched critical points it takes the pair of connecting paths with the highest mean node similarity. A dynamic program aligns the inner nodes of the two paths one to one in path order, pairing only nodes that share a label; it maximizes the number of pairs first and their total similarity second. Unpaired nodes of the longer path are dropped. Every comparison is strict, so a tie goes to the first candidate in node enumeration order.

Attributes are generalized with the structure. A numeric attribute becomes an interval with three values: minimum, maximum and centre. A categorical attribute survives only if it agrees across all samples, and label lists are replaced by their intersection.

The procedure consumes every sample it is given, in the

order the graph database returns them. The current code sorts by a random identifier assigned to each image at submission, so the order is arbitrary with respect to the samples and differs between formation runs; the order behind the baseline alphabet was not recorded. Section IV-D reports how far the result moved across three such runs. The number of samples per concept attractor was chosen during authoring by one criterion: further samples no longer changed the concept attractor. The procedure follows the energy model of concept formation in our earlier work [26]; its invariant formulation, with concept formation as hypergraph attractor dynamics, is in [27].

D. Training Data and Sample-Selection Policy

The alphabet holds 13 concept attractors for 10 digit classes. Classes 1, 2 and 4 have two each — `1_1` and `1_3`, `2_1` and `2_2`, `4_1` and `4_2` — because these digits are written in two topologically different styles that no reduction turns into one another; the other seven classes have one each. Sizes run from 3 to 12 nodes: the largest is `2_2` with 12, the smallest `1_3` with 3; the single concept attractor of digit 7, `7_1`, has 5.

The 13 concept attractors were formed from 76 unique MNIST [28] training samples, 3 to 9 each. The count per concept attractor follows from the criterion in Section III-C; no quota was fixed in advance. Each sample carries about ten augmented variants — a small random rotation and translation — giving 805 training images.

We chose the samples by hand. They are not a random draw from the training split; the criterion was the least distorted writings, with no gaps in the contour. The choice rests on an assumption: an instance of a class that a person reads correctly already carries all the structural and topological features of that class. Formation then needs a small number of structurally complete writings rather than a representative sample of the distribution. Variation enters through augmentation — the chosen samples fix the structure, and their augmented variants set the bounds of the parameter intervals.

The limits of this assumption are only partly known. Section IV-D reports what we measured about the choice and order of samples and what we did not.

E. Classification by Graph Edit Distance

Classification matches the image graph against every concept attractor in the alphabet. A complexity pre-filter runs first: the complexity of a graph is $\kappa = |V| + |E|$, and concept attractors more complex than the image graph are dropped before any distance is computed.

GED is the minimum total cost of the insertions, deletions and substitutions of nodes and edges that turn one graph into another. Exact GED is NP-hard, so practical systems approximate it [13]. We use an anytime solver with a deadline of 5 s per comparison, which returns the best edit path found by then. Substitution costs follow a monotone ladder: NO_COST $= 0$, MINOR $= 0.25$, GENERAL $= 0.5$, SEVERE $= 0.8$, NO_MATCH $= 1.0$ and IMPOSSIBLE $= 6.0$. IMPOSSIBLE is assigned to incompatible node types; its cost blocks the substitution in practice. Inserting a node costs IMPOSSIBLE, so a concept attractor node without an image counterpart blocks the match; deleting a node costs MINOR. Deleting an edge costs MINOR, inserting one NO_COST, and edges themselves are not compared.

Node comparison uses 13 features: normalized x, normalized y, distance to centroid, horizontal direction, vertical direction, angle with the x axis, minimum junction angle, endpoint indicator, corner indicator, length ratio to maximum, average neighbouring vector length, neighbouring endpoint count and neighbouring junction count. Nine of them store an interval in the concept attractor node and take a diagnostic weight, (3):

$$w_f = \frac{s_f}{\Delta_f + \varepsilon_w} \tag{3}$$

where Δ_f is the interval width, $\varepsilon_w = 0.1$ smooths the denominator and $s_f = 1$. A narrow interval means the feature was stable across the training samples, so it weighs more; a wide interval barely affects the cost. The other four features are categorical and have no interval: the two directions take $s_f = 0.6$, the endpoint and corner indicators 0.4. Weights are normalised over the features present on both nodes, $\tilde{w}_f = w_f / \sum_g w_g$. A value inside the interval costs $\tilde{w}_f \, |v - c_f| / (\Delta_f / 2)$, zero at the centre c_f and $\tilde{w}_f$ at the edge. A value outside costs the full NO_MATCH penalty of 1.0, unweighted. A node's substitution cost is the sum over its features, so one out-of-range feature alone lifts it above 1.0.

The total edit cost c becomes a similarity by (4):

$$\mathrm{sim}(G_1, G_2) = 1 - \frac{c}{c + \max(\kappa_1,\, \kappa_2)} \tag{4}$$

where κ_1 and κ_2 are the complexities of the two graphs. Similarity lies between 0 and 1; normalizing by the larger graph keeps it comparable across concept attractors of different size. Selection is winner-take-all on a score corrected for complexity, (5):

$$\mathrm{score} = \mathrm{sim} + \lambda \log_2 \kappa \tag{5}$$

with $\lambda = 0.10$, a mild preference for more complex concept attractors when similarities are close. The highest score among the concept attractors that passed the pre-filter wins. If none activates, the image is marked unrecognized.

IV. Results

A. Evaluation Data and the Completeness Filter

We evaluate on the MNIST test set [28], 10,000 handwritten digit images rendered at 100×100 pixels. The completeness filter is a per-image annotation made by hand before any classification run and recorded in `mnist_all_manifest.csv` in the public repository: an image is marked complete when its stroke has no break and incomplete otherwise. The annotator saw three panels per image, the grey-scale original, the binary mask at $\theta = 110$ and the one-pixel skeleton, and marked the image complete when the digit is fully drawn with all strokes connected. The annotation uses only that structural property, not the class label and not the classifier's output. Each image carries one label, and inter-rater agreement was not measured.

TABLE I
PER-CLASS EVALUATION. "TEST" IS THE CLASS SIZE IN THE MNIST TEST SET, "ADM." THE COUNT PASSING THE COMPLETENESS FILTER AND "CLAS." THE COUNT THAT PRODUCED A CLASSIFICATION DECISION. "COV." IS THE SHARE OF THE CLASS THAT RECEIVED A CLASS LABEL. PRECISION, RECALL AND F1 ARE COMPUTED ON THE CLASSIFIED IMAGES, WITH SUPPORT-WEIGHTED TOTALS. "FULL" IS THE RECALL OVER ALL TEST IMAGES OF THE CLASS IN THE RUN WITHOUT THE FILTER. THE LAST FIVE COLUMNS ARE PERCENTAGES.

Class	Test	Adm.	Clas.	Cov.	Prec.	Recall	F1	Full
0	980	830	830	84.4	99.87	91.93	95.73	81.33
1	1,135	1,129	1,122	98.9	93.69	96.52	95.08	95.68
2	1,032	956	955	92.1	92.28	77.59	84.30	72.87
3	1,010	969	968	95.8	91.78	95.76	93.73	94.36
4	982	953	950	96.7	92.15	84.00	87.89	83.10
5	892	796	796	89.1	93.97	91.96	92.95	88.79
6	958	742	741	75.7	84.88	92.44	88.50	74.01
7	1,028	1,014	1,011	98.3	82.01	95.15	88.10	94.84
8	974	582	581	58.2	98.74	94.49	96.57	62.11
9	1,009	736	731	72.0	92.85	92.34	92.59	72.94
Total	10,000	8,707	8,685	86.4	91.92	91.13	91.34	82.21

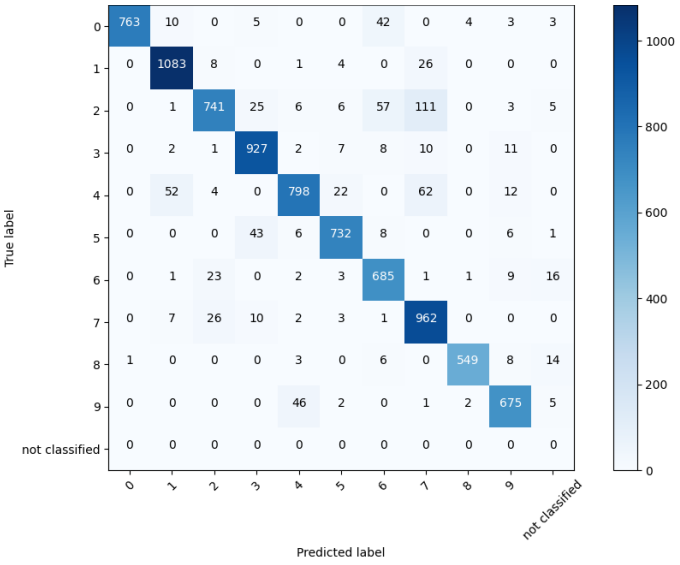


Fig. 3. Confusion matrix over the 8,685 classified images, taken from the run artefacts. Rows are true classes, columns predicted classes; the last column holds images that activated no concept, and the last row is empty by construction.

Other revisions of the manifest list 8,704 and 8,708 complete images, so the labels of a few images are unstable. The system does not detect completeness itself. The repository holds a heuristic detector built on connected components, erosion and skeleton endpoints; its agreement with the manual labels was not measured, and the evaluation does not use it. A broken stroke violates the assumption the graph representation of Section III rests on, and the graph built from such an image describes a different figure. Recognising broken structures needs a separate mechanism, associative recognition, which the system does not implement.

The filter admits 8,707 of the 10,000 test images and excludes 1,293, or 12.9%. The accuracy reported first is therefore conditional on the filter and does not transfer to full MNIST; Section IV-B adds the figures over the full test set. The filter also acts unevenly across classes (Table I). Class 1 keeps 1,129 of its 1,135 test images (99.5%), while class 8 keeps 582 of 974 (59.8%). Per-class figures therefore rest on samples of very different size, and class 8 has the narrowest basis.

A further 22 admitted images (0.25%) failed graph construction: skeletonization produced no connected graph at any threshold of the retry ladder (Section III-B). We count these as pipeline errors. A classification decision was produced for 8,685 images.

B. Classification Results

On the 8,685 classified images the system reaches 91.13% accuracy, 91.92% precision, 91.13% recall and 91.34% F1. The three averaged figures are weighted by class support; the unweighted macro averages over the ten classes are 92.22%, 91.22% and 91.54%.

Per-class figures spread much wider than the totals (Table I). Precision is highest for class 0 (99.87%) and class 8 (98.74%), whose concepts attract almost nothing from other classes. Precision is lowest for class 7 (82.01%) and class 6 (84.88%), which take on images belonging to neighbouring classes. Recall orders the classes differently, with class 2 (77.59%) and class 4 (84.00%) at the bottom.

The system made 770 wrong decisions: 726 assigned the image to a wrong class, and in 44 no concept activated and the image was left unrecognised, mostly in class 6 (16) and class 8 (14).

Counted against all 10,000 test images, three mechanisms reject an image: the filter (1,293 images, 12.93%), failed graph construction (22, 0.22%) and no activated concept (44, 0.44%). Only the last two are automatic. The reject rate is 13.59% and the coverage 86.41%: 8,641 images receive a class. The selective risk, the error rate among those 8,641, is 8.40% (726 errors). With every rejected image counted as an error, accuracy over the full test set is 79.15% (7,915 of 10,000). Coverage runs from 58.2% for class 8 to 98.9% for class 1 (Table I), and the error rate among the labelled images of a class from 3.2% for class 8 to 22.0% for class 2.

A separate run submitted all 10,000 test images with no filter, using the same 13 concept attractors and the same parameters. Graph construction failed for 58 images, 54 activated no concept and 1,667 received a wrong class, so 8,221 are correct: 82.21% of the full test set. On the 9,942 classified images accuracy is 82.69%, precision 84.57%, recall 82.69% and F1 82.83%. The last column of Table I gives the per-class recall of this run over all test images; class 8 falls to 62.11%. The manifest revision of this run lists 1,295 incomplete images, and the run is correct for 310 of them, or 23.94%. The run used a later code revision, which softens the out-of-range penalty of Section III-E for narrow intervals. On the images annotated complete it scores 90.89%, against 90.90% for the baseline run counted the same way.

Errors concentrate on a few pairs (Fig. 3). Written as true

TABLE II
DECISION TRACE FOR MNIST_TEST_2_00766.

Concept	Nodes	κ	Matching outcome	Sim.	Score
2_2	12	24	removed by the pre-filter, $24 > 21$	—	—
7_1	5	9	all five nodes substituted inside the intervals, costs 0.04–0.39	0.8317	1.149
2_1	7	13	three nodes with one out-of-range feature each; node costs 1.01 / 1.10 / 1.30	0.7300	1.100

class → predicted class, the most frequent pairs are 2→7 with 111 cases, 4→7 with 62, 2→6 with 57, 4→1 with 52, 9→4 with 46, 5→3 with 43, 0→6 with 42, 1→7 with 26, 7→2 with 26, and 2→3 with 25. These ten pairs cover 490 of the 726 wrong-class assignments, or 67.5%. Two classes produce nearly half of all errors: class 2 with 214 and class 4 with 152.

C. Decision Trace

Every decision decomposes into values a reader can check. We show this on image mnist_test_2_00766, true class 2, predicted 7. We chose a misclassification: an explanation has to hold for a wrong decision as well as for a correct one. Fig. 4 shows the construction of the graph and the two competing concepts; Table II gives the decision trace.

The image graph has complexity 21. The complexity pre-filter runs first. The richer concept of digit 2, 2_2, has 12 nodes and complexity 24, so the condition $24 > 21$ removes it before any comparison. The impoverished input is visible in the image: binarization lost the lower loop of the 2, the closed structure became an open spur, and the graph holds neither a cycle nor an intersection point.

Two concepts then compete. Concept 7_1 has five nodes, every one of them substituted within the learned intervals of the concept, at substitution costs between 0.04 and 0.39, and reaches similarity 0.8317. Concept 2_1 has seven nodes and matches the input structurally, but three of its nodes each carry one feature outside its interval, and each takes the full 1.0 penalty: the StartPoint on distance_to_centroid (0.56 against $[0.69, 1.00]$), a CornerPoint on normalized_x (0.20 against $[-0.70, 0.10]$) and a horizontal Vector node on length_ratio_to_max (0.31 against $[0.43, 1.00]$). With their in-range features added, the three node substitution costs are 1.01, 1.10 and 1.30. Its similarity is 0.7300.

Complexity is $\kappa = 9$ for 7_1 and $\kappa = 13$ for 2_1, so the prior of (5) favours 2_1 by 0.053 and does not close the gap. Winner-take-all on the score picks 7_1: 1.149 against 1.100.

The mechanism holds beyond this one image. Across all 111 2→7 errors, 2_1 entered the comparison every time and lost on similarity every time, by a mean margin of 0.14. In none of the 111 cases did a concept of digit 2 win on similarity and then lose after the complexity prior was applied.

D. Sensitivity and Ablations

No single-factor ablation was run on the baseline configuration; the results below come from other runs.

Three back-to-back retrains on later revisions of the code and of the training corpus (869 images from 85 samples) each consumed the samples in a different order. All three reproduced the node count of every concept attractor in the baseline alphabet. Accuracy on the classified images was 90.68%, 89.93% and 90.93%. In an offline study, ten shuffled orders gave 7_1 and 8_1 the same labelled topology every time; 2_2 kept 12 nodes, and 64% of its order pairs had identical topology. We did not form concepts from a random draw of the training split, did not vary the number of concept attractors, and did not vary augmentation. The effect of those three choices on accuracy is unknown.

The previous configuration scored 85.80% under the same protocol. It used the cost ladder 0, 0.65, 0.75, 1.0, 1.5, 10, $\varepsilon_w = 1.0$, 14 features and a 15 s GED deadline, and it preceded one change to skeletonization and the re-formation of five concept attractors. The gain of 5.33 points cannot be split among these six changes. The later code revision of Section IV-B softens the out-of-range penalty and leaves accuracy on complete images within 0.02 points, though it carries other code changes as well. On an earlier configuration and a 3,000-image sample, removing the complexity prior of (5) lowered accuracy by about 3 points, to 72.00%. For the 111 2→7 errors, lifting the complexity pre-filter changed no decision (Section V). We did not run the evaluation set without the pre-filter and did not test a soft pre-filter.

E. Inference Cost

An image is compared against at most 13 concepts, each comparison under a 5 s deadline, so the theoretical upper bound is 65 s per image. The measured times stay far below it.

We measured end-to-end time over the whole evaluation set on one Apple M1 Pro workstation (10 cores, 32 GB) running the pipeline in 26 Docker containers: a connector, four skeletonization, four contour analysis and sixteen classification instances, and the concept creator. The clock runs from the first HTTP submission to the last result and includes database writes and message-queue hops. The three runs of Section IV-D took 1,279 s, 1,431 s and 2,018 s for 8,704 to 8,708 images, which is 147, 164 and 232 ms per image. These are throughput figures with the stages running in parallel; the latency of a single image was not recorded. Each run was slower than the one before. Every run leaves its test graphs in the database, which held 514k nodes after the third; that growth is the likely cause, but we did not isolate it. Forming the 13 concept attractors took 528 s, 746 s and 966 s.

Per-stage times exist only for the 770 misclassified images of the baseline run, because a run stores profiling records only for those. Their medians are 52.8 ms for skeletonization (maximum 533 ms), 211.5 ms for contour analysis (maximum 1,905 ms) and 167.1 ms for the classification stage, meaning all GED comparisons for one image together (99th percentile

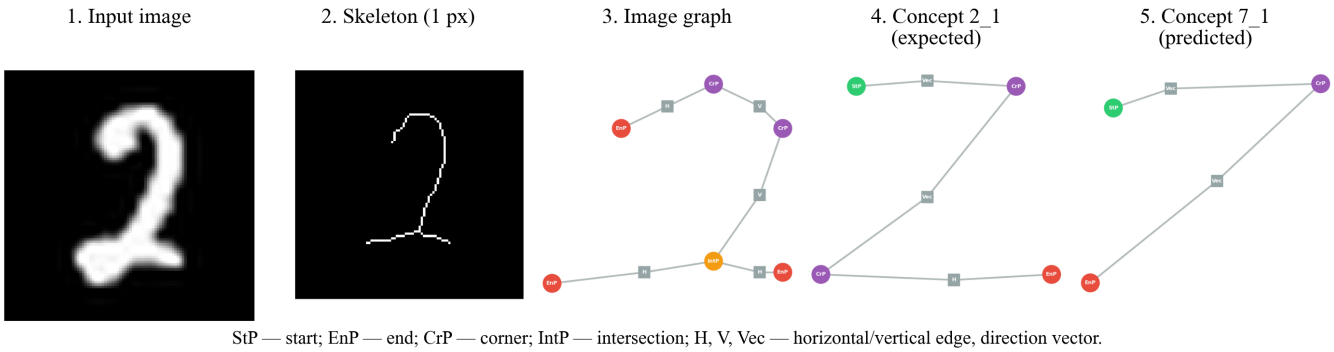


Fig. 4. Decision trace for `mnist_test_2_00766`, in five panels: the input image; the 1-pixel skeleton; the image graph with Point and Vector nodes; concept `2_1`, expected; concept `7_1`, predicted. The lower loop of the digit is already absent in the skeleton, so the image graph carries neither a cycle nor an intersection point.

about 0.84 s, maximum 1,414 ms). No profiled image came close to the 5 s deadline of a single comparison, so on those images the solver finished its search and the reported distances are completed results. The subset holds no correctly classified image, so these medians are not a typical-case estimate.

Three quantities were not measured: the time of an individual GED comparison, memory use, and the growth of inference time with the number of concept attractors. The number of comparisons per image is at most the number of concept attractors that pass the pre-filter.

V. Discussion

The explanation here is the decision object itself. No second model approximates the classifier from outside. To see why an image received a class, a reader opens the image graph and reads the learned intervals of the winning concept attractor and the cost charged for each feature. The trace in Section IV-C runs the whole chain, from the complexity pre-filter down to one out-of-range value, `normalized_x` at 0.20 against the interval $[-0.70, 0.10]$. Post-hoc attribution methods build their explanation after the fact by perturbing the input [1], [2], and that construction can be manipulated [3]. Decisions with consequences are better taken by models interpretable by construction [4].

The same property applies to the failures. Binarization loses the lower loop of the digit, so the image graph carries no intersection point, and the pre-filter removes `2_2` in 86 of the 111 2→7 cases. The pre-filter is not the cause: in a control run with it lifted, `2_2` won none of the 111, because an intersection-point condition blocks its match to a graph without an intersection. The error has two causes: the lower loop lost at binarization, and the full interval penalties on `2_1` against the in-interval match of the five-node `7_1`. The error can be named and its prevalence checked over all 111 cases.

Traceability defines the application niche: tasks where the answer has to be auditable. Verification of handwritten marks in document workflows, inspection of industrial marking, and tutoring systems that must show a learner the reason for a grade share the property that a human resolves every disputed decision. A decision record of the named concept, its node substitutions and its per-feature costs goes into that review without a separate explanation module. For a dependability audience the property on offer is auditability: every decision leaves a record that a person can check against the input without re-running the system. The small training set also lowers the cost of starting on a new set of marks: one concept attractor forms from 3–9 samples and their augmented variants (Section III-D). An earlier prototype reported 82.44% on six classes [26]; the two numbers are not comparable, since the class set, the test set size and the formation procedure all differ.

Several limitations bound the result. The headline metric is conditional on the completeness filter, a manual annotation with one label per image and no measured inter-rater agreement, which excludes 12.9% of the test set unevenly across classes. The training samples were chosen on the same completeness criterion as the test filter, so the evaluation measures the system inside the regime it was built for, and the exclusion does the most work in the classes with the lowest admission rates (8, 9 and 6). The 76 training samples were selected by hand. A different sample order moved accuracy by about one point (Section IV-D), but random selection, the number of concept attractors and augmentation were not varied, so part of the accuracy may come from a fortunate choice. We ran no non-neural structural baseline: no classifier on contour or skeleton descriptors, no template matcher, and no nearest-neighbour GED against the unreduced training graphs. The last one would show what the reduction to 13 concept attractors contributes. The paper therefore supports no claim that concept attractors outperform other structural methods. No single-factor ablation of the edit costs, the feature weights, the complexity prior or the pre-filter was run on the baseline configuration. Inference time is measured end to end, but per-stage records cover only misclassified images, and per-comparison time, memory use and scaling with the number of concept attractors were not measured. The evaluation covers MNIST only. We evaluated no second dataset and none of the application tasks named above, so the paper does not establish generalization beyond handwritten digits or dependability in

those workflows. Tolerance to rotation is limited to the small range covered by augmentation. The contour representation discards texture and gradients, which confines the method to silhouette and stroke images. Broken contours need associative recognition, which the system does not have.

VI. Conclusion

We formed an alphabet of 13 concept attractors for ten digit classes by structural reduction of 76 hand-selected training samples and their augmented variants, with no backpropagation anywhere in the procedure. Classification by graph edit distance with interval-based feature weighting recognises 8,685 complete-contour MNIST images at 91.13% accuracy, 91.92% precision, 91.13% recall and 91.34% F1. The evaluation is stated with its filter: 12.9% of the test set is excluded. Counted over all 10,000 test images the system assigns a class to 86.41% at a selective risk of 8.40%, and without the filter it classifies 82.21% correctly. Each decision decomposes into a named concept, a list of node substitutions and a per-feature cost, and the dominant 2→7 error was traced through that decomposition over all 111 cases. End-to-end inference took 147 to 232 ms per image, and on the profiled images the 5 s deadline of a single comparison was not reached. Work ahead covers associative recognition for broken contours, a nearest-neighbour GED baseline on the unreduced training graphs, single-factor ablations of the matching costs, and formation from randomly drawn training samples. The source code and the run artefacts are public at https://github.com/NaturalAGI-Lab/NaturalAGI.

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